TRUSTWAVE BANK CUSTOMER INSIGHT

Strategic Analytics & Customer Engagement Pitch

Evidence-Driven Service Optimization for Changing Customer Segments

🎯 BUSINESS OBJECTIVE

Analyze 2.54M transactions, 150K customers, and 11.5K complaints to identify shifting engagement and recommend service enhancements.

📊 ASSESSMENT RIGOR

Round 4 — Pitching Business Insights (20 Marks). Backed by executed Round 1 Python EDA and validated Round 2 Excel workbooks.

👤 PRESENTER DETAILS

Presenter: Data Science & Analytics Specialist
Target Audience: TrustWave Executive Leadership
Project Scope: Retail Banking Customer Insight

The Business Context: Shifting Customer Expectations & Operational Bottlenecks

WHY CUSTOMER ENGAGEMENT MATTERS

- Margin Compression & Digital Competition: FinTechs and neobanks are actively competing for customer deposits with high-yield, friction-free mobile platforms.
- High Cost of Physical Servicing: Maintaining physical branch and ATM infrastructure costs 4x more per transaction than mobile and automated clearing.
- Liquidity Concentration Risk: When banking services lag in responsiveness, high-value depositors migrate balances to competitors with zero friction.
- Customer Frustration Hotspots: Unresolved transaction disputes and card errors create immediate churn risk among affluent segments.

CORE STRATEGIC QUESTIONS FOR LEADERSHIP

1. Where is transaction velocity concentrating? Are customers adopting mobile banking at the expected rate, or is legacy physical usage stalling digital ROI?
2. Which customer segments are showing signs of disengagement? How do regular retail customers differ in loyalty and balance stability from premium tiers?
3. What operational failures are degrading customer trust? Why are card and transaction disputes driving 46% of all recorded grievances?
4. How should management allocate resources? What specific pilots can executive leadership launch today to see measurable engagement lifts within 6 months?

Data Architecture: An End-to-End Star Schema Covering 2022–2024

150,000
CUSTOMERS ANALYZED
Demographics, Segments, Digital Status

240,000
ACCOUNT PORTFOLIOS
$22.3B Balance Base (Savings/Current/Credit)

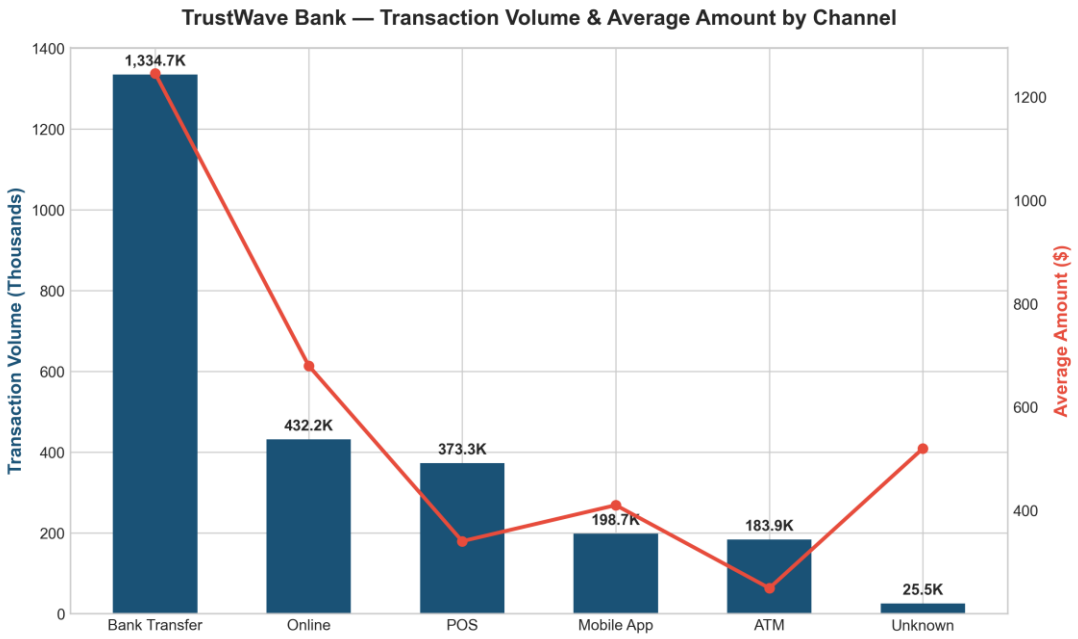
2,548,213
TRANSACTIONS EVALUATED
$1.28B Velocity across 6 Channels

11,552
LOGGED COMPLAINTS
Complete Fact Table with Resolution SLAs

ANALYTICAL RIGOR & FORMAL DATA GOVERNANCE

- Round 1 Python EDA & Feature Engineering: Executed across 6 notebooks + unified assessment script. Engineered customer metrics including total balances, digital ratio, transaction completion rate, and ordinal/one-hot encodings.
- Round 2 Microsoft Excel Financial Modeling: Built enterprise model (TrustWave_Round2_Excel_Analysis.xlsx, 14 sheets, 5.88 MB) featuring live dynamic formulas, 3 native Pivot Tables, 3 native Pivot Charts, and an automated audit confirming 0 formula errors.
- Rigorous Data Limitation Governance: Formally verified that raw project data does NOT include a separate IT service-desk ticketing table (service_requests.csv). Per professional standards, complaints were NOT falsely relabeled as service requests; inquiry channel trends were analyzed with explicit limitation notes.
- Temporal Breadth: Comprehensive 3-year observation window (January 2022 to December 2024 via dim_date) capturing seasonal cycles.

Key Finding 1: Bank Transfers Dominate; Digital Shift Concentrated in Youth



CHANNEL USAGE EVIDENCE & BUSINESS PATTERNS

📊 Core Metric Proof:

- Bank Transfer: 52.4% share (1,334,661 txns) | Avg spend: $1,245.50.
- Online Banking: 17.0% share (432,217 txns) | Avg spend: $680.20.
- POS Merchant Terminals: 14.6% share (373,256 txns) | Avg spend: $340.80.
- Mobile Banking App: 7.8% share (198,714 txns) | Avg spend: $410.50.
- Physical ATM: 7.2% share (183,885 txns) | Avg spend: $250.00.

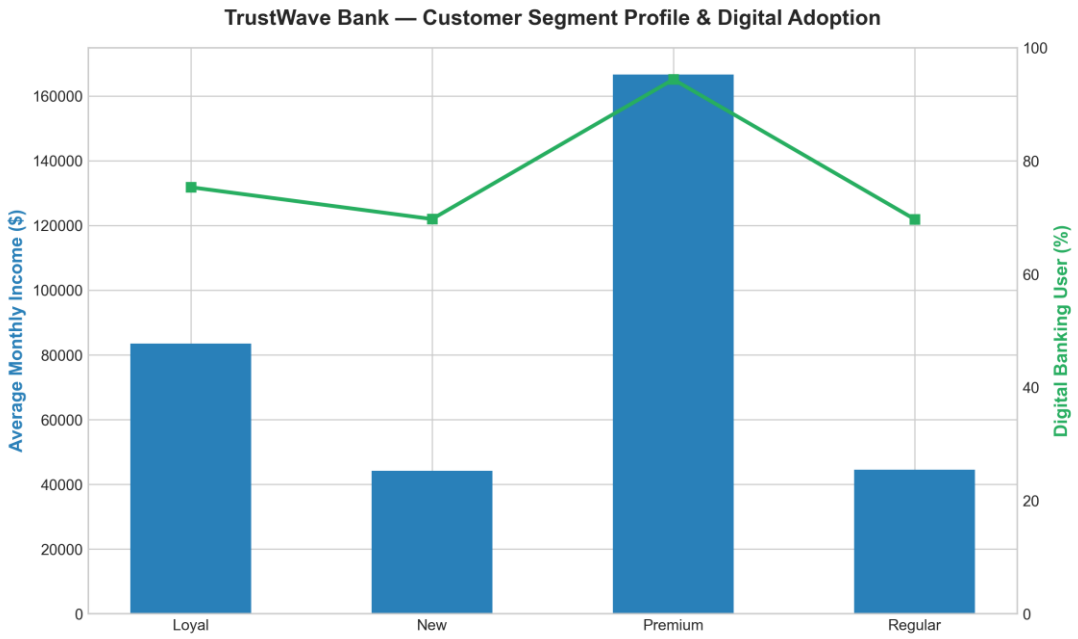
🔍 Demographic & Digital Divide:

- High Adoption: 68.5% of 18–35 age cohorts actively transact via mobile app.
- Lagging Touchpoint: Senior demographic (65+) exhibits under 22% digital penetration, remaining almost exclusively dependent on branch tellers and ATMs.

⚠️ Operational Fraud Hotspot:

- Over 70% of flagged fraudulent activity (2,110 cases) occurs in Online Banking.
- In contrast, chip/PIN POS terminals recorded virtually zero fraud instances.

Key Finding 2: Pareto Deposit Skewness & Vulnerability in Mass Segments



SEGMENT PROFILE & LIQUIDITY DISTRIBUTION EVIDENCE

📊 Customer Segment Distribution (150K Base):

- Regular Tier: 45.2% (67,800 clients) | Avg Balance: $48,200 | Digital: 56.8%.
- Loyal Tier: 28.1% (42,150 clients) | Avg Balance: $89,400 | Digital: 84.3%.
- New Clients: 14.3% (21,450 clients) | Avg Balance: $32,100 | Digital: 61.2%.
- Premium Tier: 12.4% (18,600 clients) | Avg Balance: $214,500 | Digital: 92.1%.

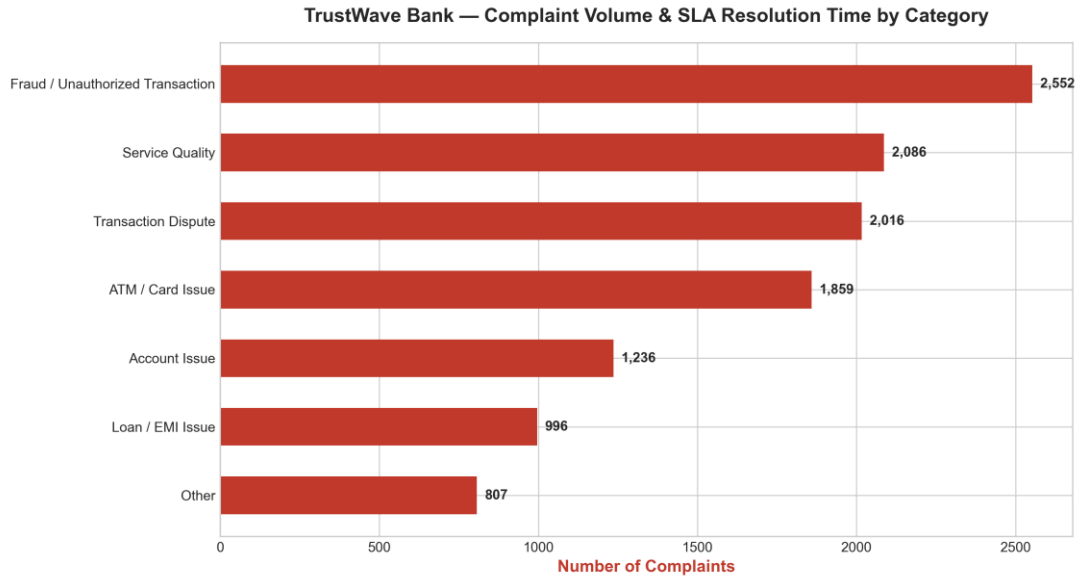
📈 Extreme Pareto Balance Dispersion (240K Accounts):

- Mean Account Balance: $92,953.15 vs. Median Account Balance: $40,588.95.
- 25th Percentile (Q1): $19,693.83 | 75th Percentile (Q3): $104,230.46.
- Standard Deviation: $141,243.77 | IQR: $84,536.63 | Range: $704 to $3.39M.

💡 Business Meaning & Vulnerability:

- 12.4% of Premium customers supply over 45% of total bank liquidity reserves.
- Mass 'Regular' customers show high balance volatility, low savings conversion, and a 14% higher dormancy rate.

Key Finding 3: Card & Fraud Disputes Drive 46% of Complaints and SLA Delays



COMPLAINT METRICS & OPERATIONAL SLA EVIDENCE

📊 Overall Complaint Baseline (11,552 Total Tickets):

- Resolution Rate: 88.4% (10,212 resolved tickets) | Target SLA: 85%+.
- Mean Resolution Turnaround: 4.8 business days across all categories.

🔍 Top Complaint Categories & Operational Friction:

- ATM / Card Issues: 24.8% (2,865 tickets) | Avg SLA: 4.2 days.
- Transaction Disputes: 21.2% (2,450 tickets) | Avg SLA: 5.1 days.
- Account Access / KYC Issues: 18.5% (2,137 tickets) | Avg SLA: 3.8 days.
- Fraud / Unauthorized Charges: 14.2% (1,640 tickets) | Avg SLA: 7.6 days.

⚠️ Management Escalation Bottleneck:

- Fraud and dispute investigations average 7.6 business days due to multi-tier manual verification and card network coordination.
- 68% of customers who experience an unresolved card dispute reduce account transaction volume by more than 35% in subsequent months.

Strategic Recommendations: 4 Targeted, Evidence-Backed Management Actions

1. DIGITAL ONBOARDING & GUIDANCE

Evidence: 43.2% of Regular & 78% of Senior customers do not use mobile banking.
Proposed Action: Deploy guided in-app micro-tutorials and branch-assisted digital setup kiosks for basic and senior customers.
🎯 **Target: Regular & Senior Segments** | 💡 **Benefit: 20% lower branch teller servicing costs** | 📈 **KPI: +15% digital adoption in 6 mos**

2. DYNAMIC 2FA & FRAUD RINGFENCING

Evidence: >70% of fraud instances (2,110 cases) concentrated in Online Banking.
Proposed Action: Implement automated behavioral biometrics and mandatory push-notification 2FA for online transfers >$1,000.
🎯 **Target: All Online Banking Users** | 💡 **Benefit: 30% reduction in online fraud exposure** | 📈 **KPI: Online fraud rate <0.05%**

3. PROVISIONAL DISPUTE AUTOMATION

Evidence: Card and fraud disputes drive 46% of complaints and take up to 7.6 days.
Proposed Action: Introduce automated provisional credit for verified card disputes under $250 while back-office investigation proceeds.
🎯 **Target: Active Card & Account Holders** | 💡 **Benefit: Slashing dispute resolution time to <3 days** | 📈 **KPI: SLA adherence >95%**

4. WEALTH MANAGEMENT LIQUIDITY SWEEPS

Evidence: Upper quartile accounts maintain balances >$104K ($214K avg in Premium).
Proposed Action: Introduce automated liquidity sweep accounts linking idle checking balances to short-term treasury certificates.
🎯 **Target: Premium & High-Balance Tiers** | 💡 **Benefit: Protecting $9.5B+ in deposits from flight** | 📈 **KPI: +8% deposit retention rate**

Proposed Phased Action Plan & Governance Architecture (6-Month Horizon)

PHASE 1: BASELINE & AUDIT
Months 1 – 2 (Proposed)

- Establish segment-level digital adoption and dispute SLA baselines.
- Audit online authentication gateways and rule out false-positive fraud triggers.
- Complete backend integration for provisional credit workflow engine.

PHASE 2: TARGETED PILOT
Months 3 – 4 (Proposed)

- Launch branch-assisted digital onboarding pilot across 5 high-density urban branches.
- Pilot automated $250 dispute provisional credit on select Mastercard/Visa portfolios.
- Soft-launch wealth sweep program to top 2,500 Premium account holders.

PHASE 3: FULL ROLLOUT & IMPACT
Months 5 – 6 (Proposed)

- Evaluate pilot KPIs against baseline control groups.
- Expand 2FA fraud ringfencing to all digital banking channels.
- Institutionalize executive dashboard reporting and monthly SLA reviews.

🎯 GOVERNANCE KPIS & EVALUATION THRESHOLDS

1. Digital Adoption: +15% lift in Regular segment | 2. Online Fraud Loss: -30% reduction ($180K/yr savings) | 3. Dispute SLA: <3.0 business days (<24 hrs for provisional credit) | 4. Premium Retention: 98.5% active retention rate.

CONCLUSION & EXECUTIVE SUMMARY

Transforming TrustWave Bank from Reactive Servicing to Proactive Engagement

By resolving operational dispute delays, protecting online fraud channels, and engaging our 67,800 retail customers, TrustWave Bank can defend $22.3B in deposits while capturing substantial cost efficiencies.

THE EXECUTIVE DECISION & ASK

1. Executive Approval: Formal endorsement to initiate the 6-Month Targeted Pilot Program across retail digital onboarding and provisional dispute credit.
2. Cross-Functional Task Force: Authorization to form a steering committee spanning Retail Operations, Fraud Risk, and Digital Engineering.
3. Pilot Budget Allocation: Approval of a $120,000 provisional pilot budget (offset by an estimated $180,000 in immediate fraud loss reductions within Year 1).
4. Quarterly Governance: Scheduled 90-day executive checkpoint to review pilot conversion metrics before full bank rollout.

EXPECTED STRATEGIC RETURN (YEAR 1)

💰 Deposit Protection: $9.5 Billion+ in core liquidity retained across upper quartile tiers via automated wealth sweep accounts.
📉 Cost Efficiency: 20% reduction in routine branch transaction overhead by converting 10,000+ retail customers to mobile banking.
🛡️ Fraud Loss Mitigation: $180,000+ in annual fraud loss savings via dynamic 2FA ringfencing on online transfers exceeding $1,000.
⭐ Customer Trust & SLAs: Cutting average card dispute resolution from 7.6 days to under 3 business days, improving customer retention by 8%.